Exploratory Data Analysis (EDA)

Titanic passenger dataset - findings report

Data Analyst Internship - Task 5 | Elevate Labs

Prepared by: Ravi | Tools: Python, Pandas, Matplotlib, Seaborn, SciPy

1. Objective, dataset and approach

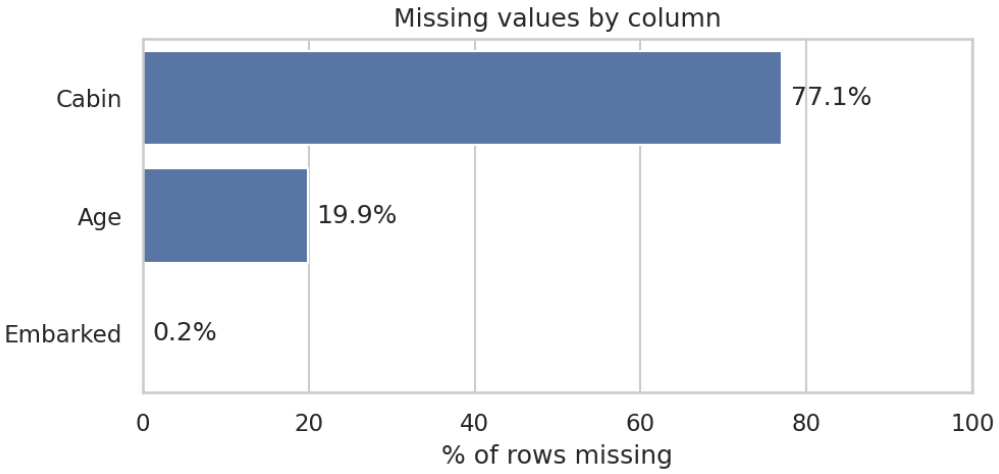
Objective: extract insights from the Titanic passenger data using visual and statistical exploration, and identify the patterns, trends and anomalies behind who survived. The full code, outputs and extra checks are in the accompanying notebook (`Titanic_EDA.ipynb`).

Item	Detail
Dataset	Titanic training set - 891 passengers, 12 columns
Target	Survived (1 = survived, 0 = died): 342 survived (38.4%), 549 died (61.6%)
Numeric columns	Age, SibSp, Parch, Fare (plus PassengerId, Survived, Pclass)
Categorical columns	Sex, Embarked, Pclass, Name, Ticket, Cabin
Method	Structure and quality check, cleaning, univariate, bivariate and multivariate analysis, statistical tests

2. Data understanding and quality

`.info()`, `.describe()` and `.value_counts()` were used to inspect types, ranges and category counts. Three columns have missing values.

Missing values



Observations

- **Cabin** is missing for 687 of 891 passengers (77.1%) - too sparse to fill in, so it is converted into a simple `HasCabin` flag.
- **Age** is missing for 177 passengers (19.9%). The gaps are not random: 136 of them are 3rd-class passengers, and passengers with a missing age survived less (29.4%) than those with a known age (40.6%).
- **Embarked** has only 2 missing values (0.2%); they are filled with the most frequent port (Southampton).

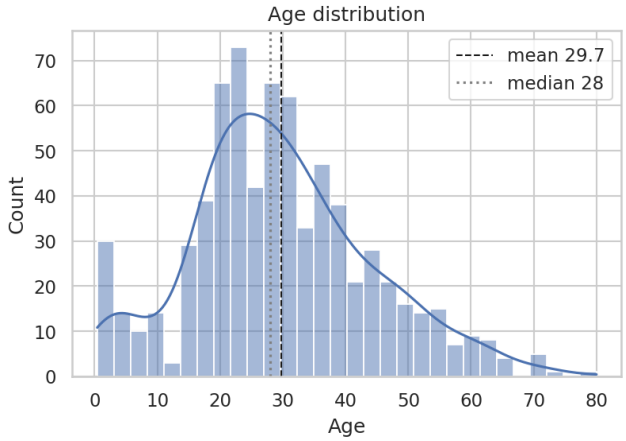
Cleaning and feature engineering

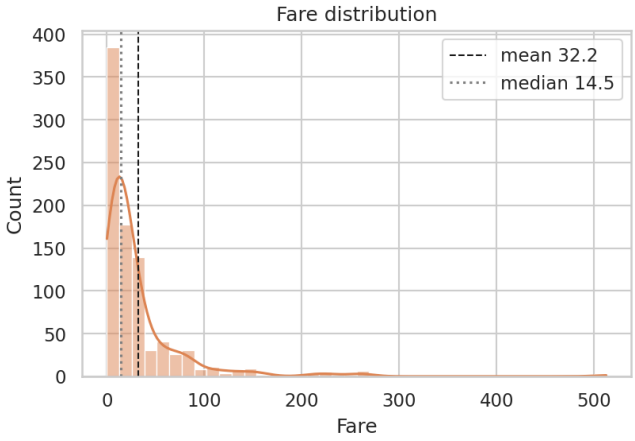
- Having a cabin recorded looks informative (66.7% survival vs 30.0% without), but it is mostly a class effect: a cabin is recorded for 81.5% of 1st-class passengers versus only 2.4% of 3rd-class passengers.

- Age is left as-is (NaN) for the EDA plots so the real distribution is not distorted; seaborn/pandas skip missing values automatically. A model-ready `Age_filled` column (median by Pclass and Sex) is saved in `titanic_clean.csv`.
- New features: `FamilySize` (SibSp + Parch + 1), `IsAlone`, `Title` (from Name), `AgeGroup`, `Outcome` (label for Survived) and later `LogFare`.

3. Univariate analysis

3.1 Histograms - Age and Fare

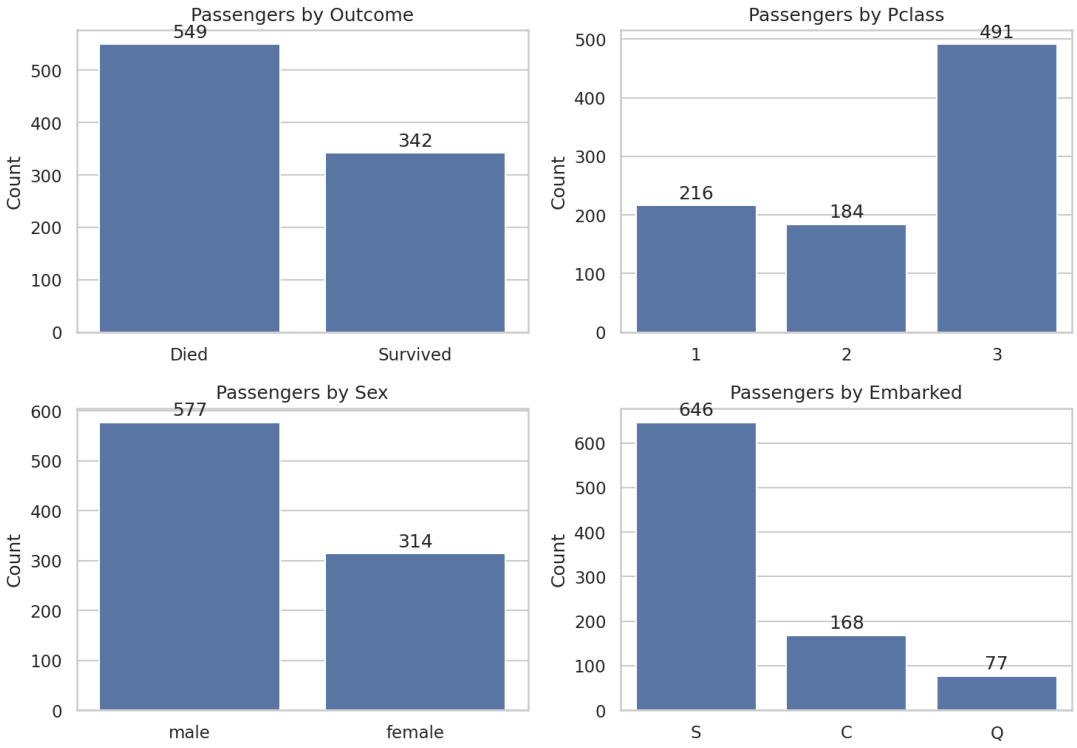




Observations

- **Age** is roughly bell-shaped (skewness 0.39) with mean 29.7 and median 28 years. There is a small bump of very young children (under 5) and a thin right tail up to 80.
- **Fare** is strongly right-skewed (skewness 4.79): median 14.45 vs mean 32.20, maximum 512.33. Most passengers paid little, a few paid a lot.

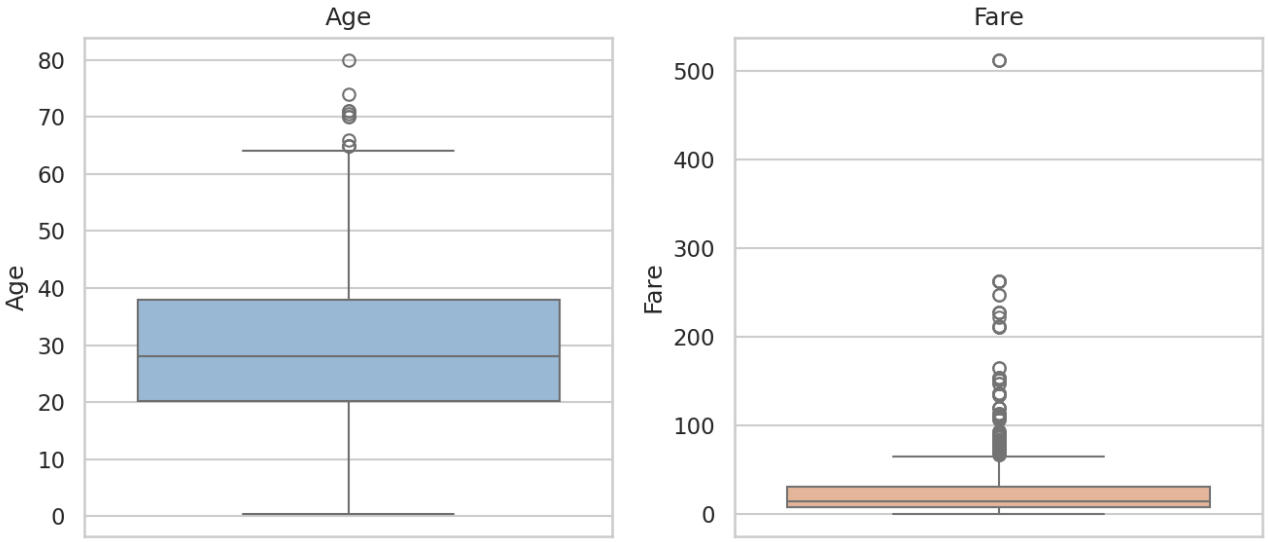
3.2 Count plots - Survived, Pclass, Sex, Embarked



Observations

- Only **342 of 891 passengers survived (38.4%)**; 549 (61.6%) died - a moderately imbalanced target.
- 3rd class is the largest group (491 passengers, 55.1%), then 1st (216, 24.2%) and 2nd (184, 20.7%).
- Men outnumber women: 577 (64.8%) vs 314 (35.2%).
- Southampton is the main port of embarkation (646 passengers, 72.5%), followed by Cherbourg (168) and Queenstown (77).

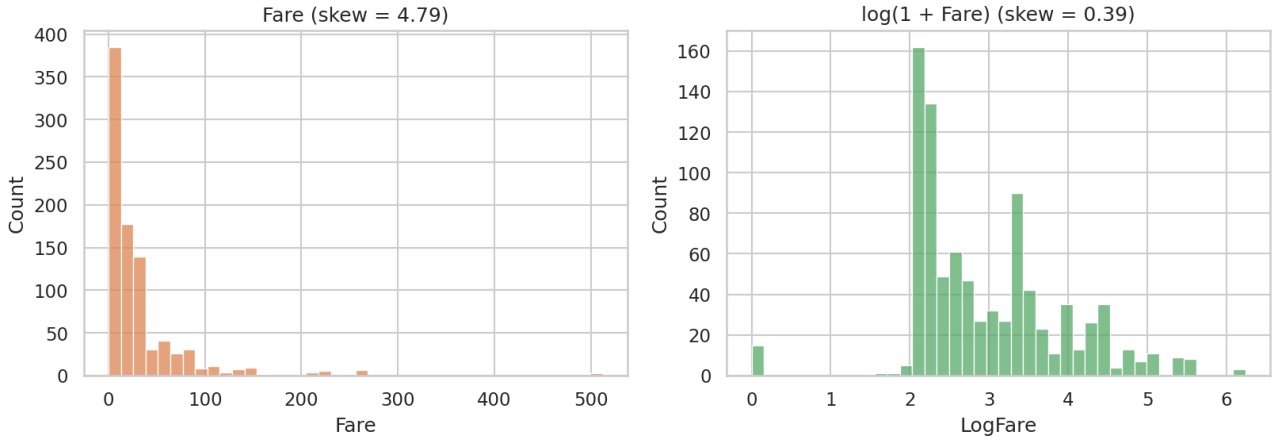
3.3 Boxplots - outlier check



Observations

- **Age**: median 28, middle 50% between 20 and 38. 11 passengers lie above the upper fence (about 65 years) - genuine elderly passengers, not data errors.
- **Fare**: the box is narrow (7.91 to 31.00) but 116 passengers (13.0%) lie above the upper fence of about 65.6, stretching up to 512.33. Fare is dominated by a long tail of expensive tickets.

3.4 Handling skew - log transform of Fare

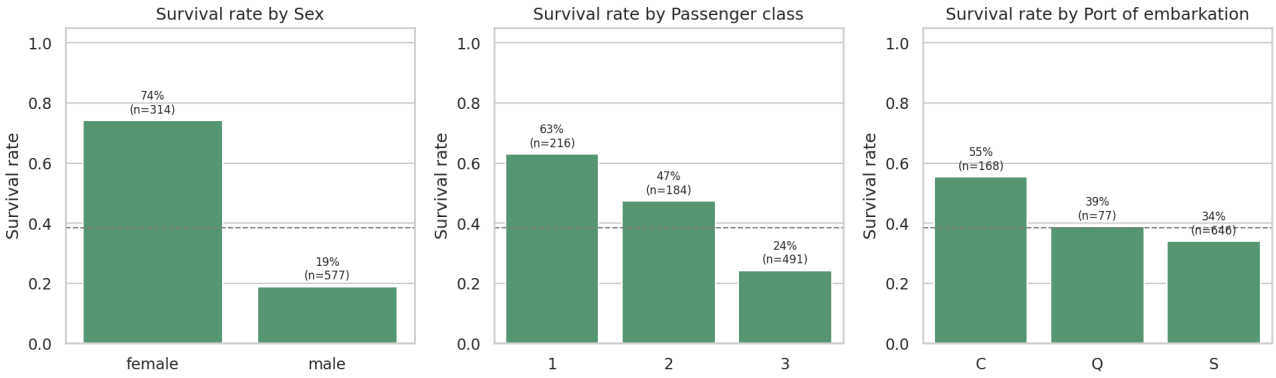


Observations

- `log(1 + Fare)` reduces the skewness from **4.79 to 0.39** and gives a near-symmetric distribution.
- `log1p` is used instead of a plain log because 15 passengers have Fare = 0.
- `LogFare` is used in the scatter, boxplot and pairplot visuals below so the extreme fares do not squash everything else.

4. Bivariate analysis

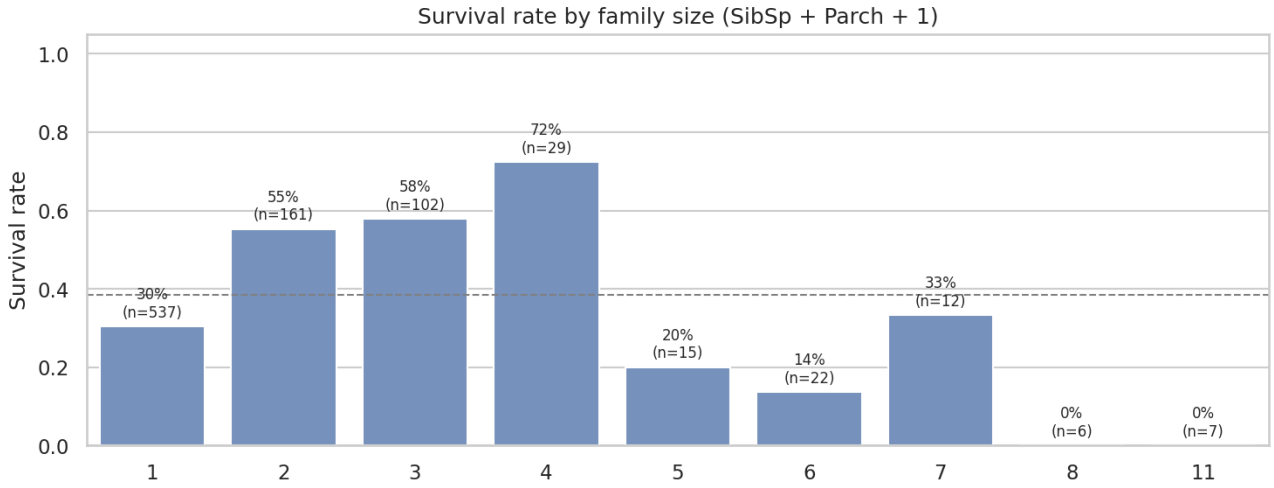
4.1 Survival rate by Sex, Pclass and Embarked



Observations

- **Sex is the strongest single factor**: 74.2% of women survived vs only 18.9% of men.
- **Survival falls with class**: 63.0% (1st), 47.3% (2nd), 24.2% (3rd).
- **Port**: Cherbourg 55.4%, Queenstown 39.0%, Southampton 33.9%. Cherbourg's advantage is partly a class effect - 51% of its passengers were 1st class vs 20% at Southampton.
- The dashed line is the overall survival rate (38.4%).

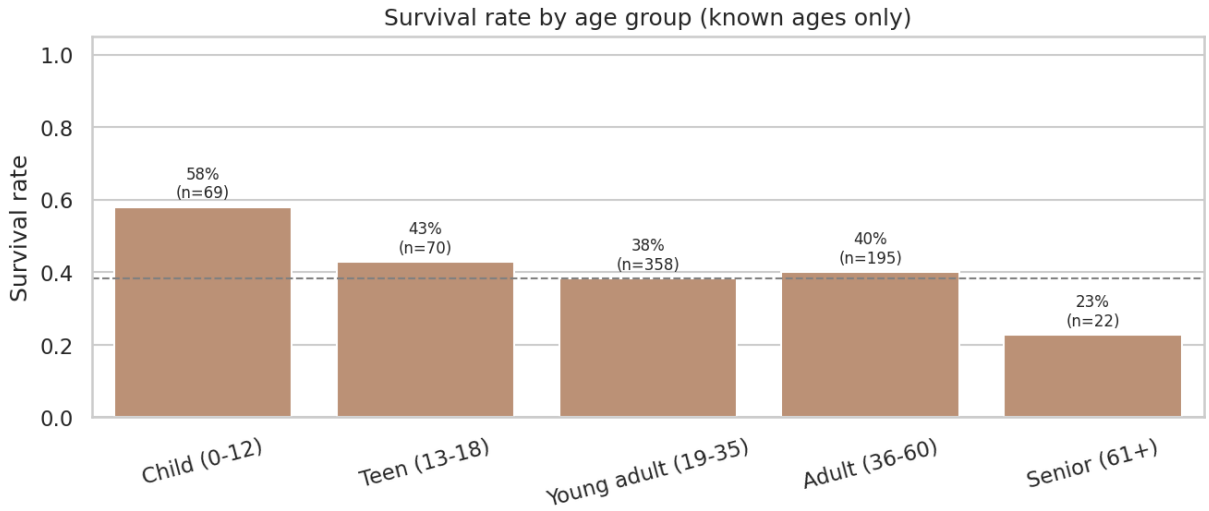
4.2 Survival rate by family size



Observations

- Passengers travelling **alone** (537 people, 60% of all) had only 30.4% survival.
- Small families of 2-4 people had the best odds: 55.3%, 57.8% and 72.4%.
- Large families (5 or more, 62 passengers) did badly - 16.1% survived, and none of the 7 passengers in the 11-person family did. Group sizes are small, so treat single bars with caution.

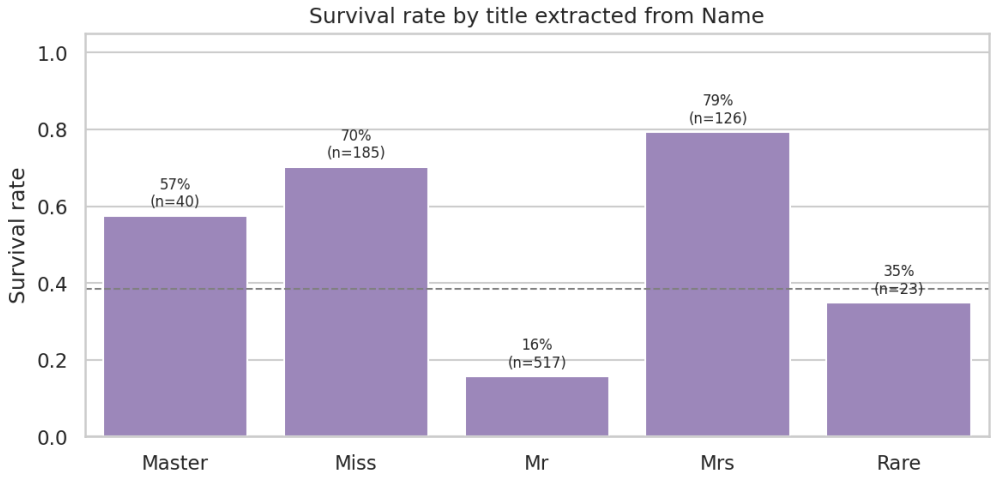
4.3 Survival rate by age group



Observations

- **Children (0-12) had the highest survival: 58.0%**, falling to 38-43% for teens and adults and just 22.7% for passengers over 60 (only 22 people).
- Age matters less than sex or class overall, but the 'children first' pattern is clearly visible.
- The 177 passengers with missing age are excluded from this plot.

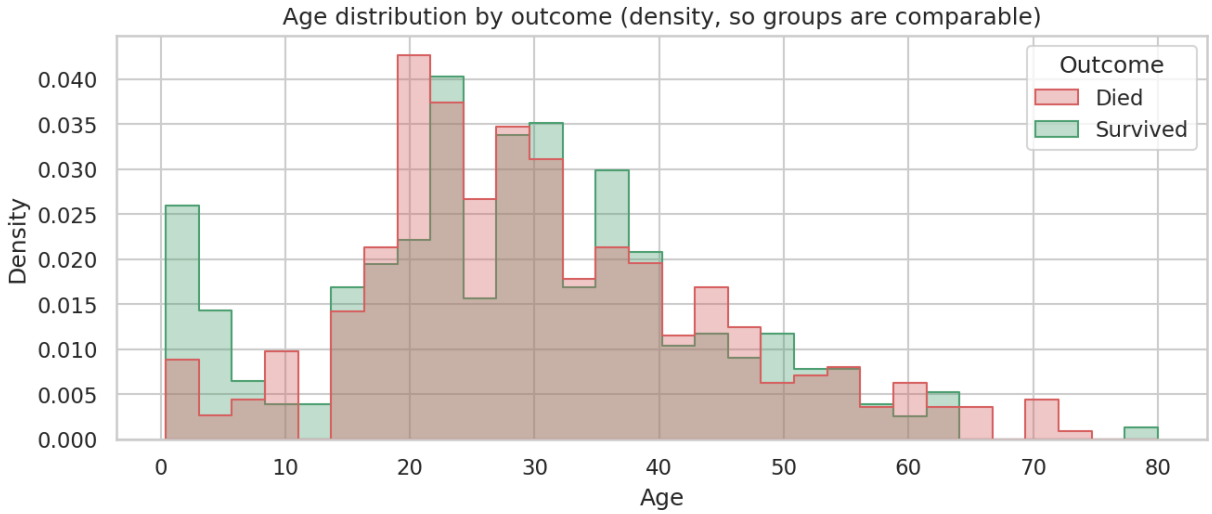
4.4 Survival rate by title



Observations

- Mr 15.7% (517 passengers), Mrs 79.4% (126), Miss 70.3% (185), Master (young boys) 57.5% (40), Rare titles 34.8% (23).
- Title captures sex, age and status in one feature: boys survived far more often than adult men (57.5% vs 15.7%), supporting a 'women and children first' pattern.

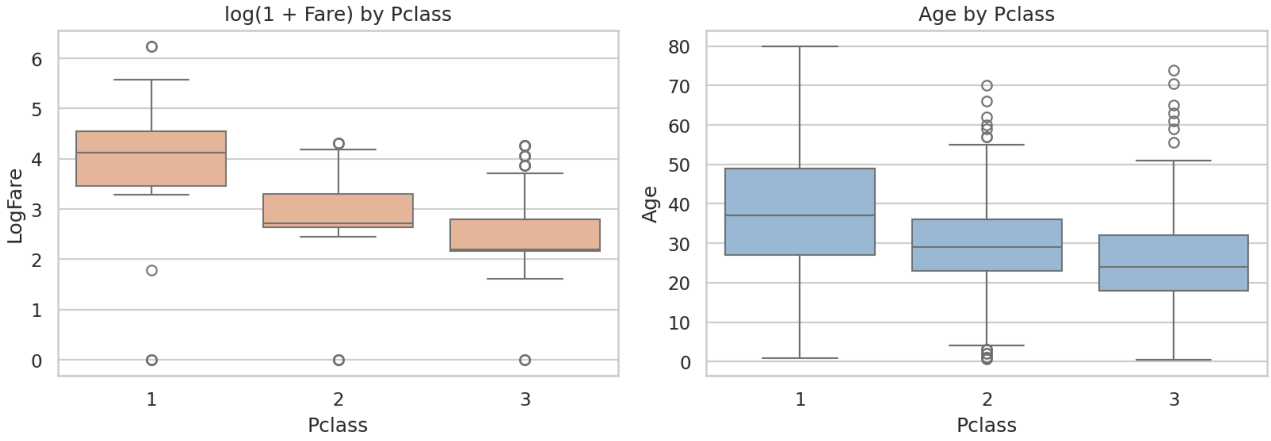
4.5 Age distribution - survivors vs non-survivors



Observations

- The two age distributions overlap heavily: median age is 28 in both groups (mean 28.3 for survivors vs 30.6 for non-survivors).
- Survivors have an extra peak among young children and fewer passengers above 60.
- A Mann-Whitney test finds no significant overall age difference (p = 0.16), so age is informative mainly at the extremes (infants and the elderly).

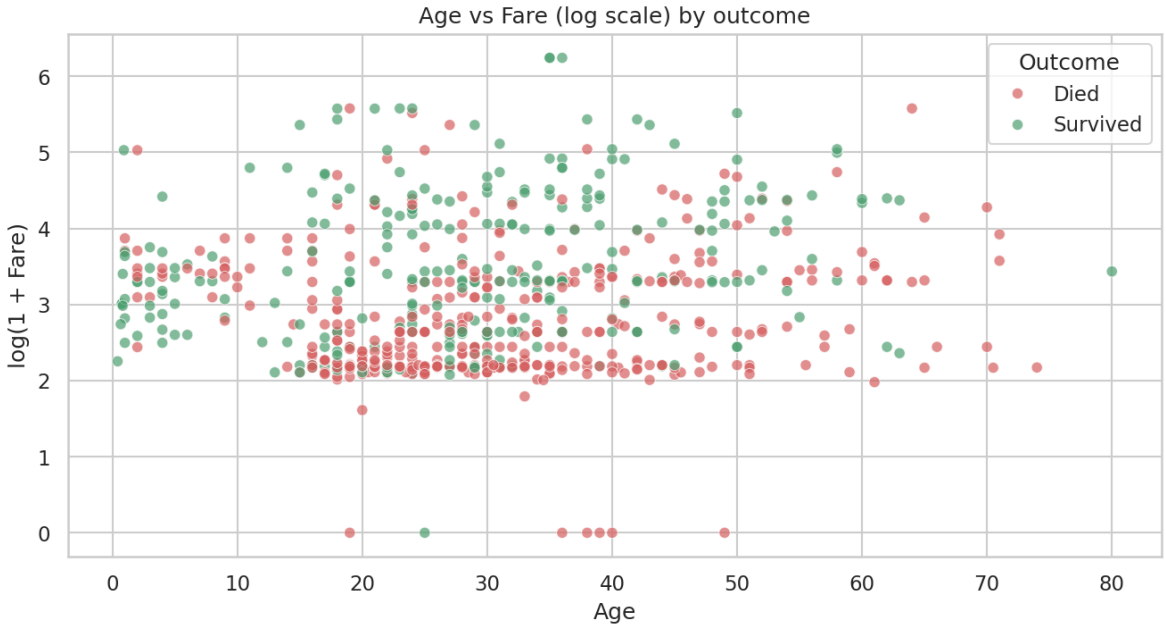
4.6 Fare and Age by passenger class



Observations

- **Median fare falls sharply with class**: 60.29 (1st), 14.25 (2nd), 8.05 (3rd); 1st class also has the widest spread.
- **Higher classes were older**: median age 37 (1st), 29 (2nd), 24 (3rd). This explains the negative Age-Pclass correlation (-0.37).
- Fare and class overlap heavily (correlation -0.55), so they carry related information.

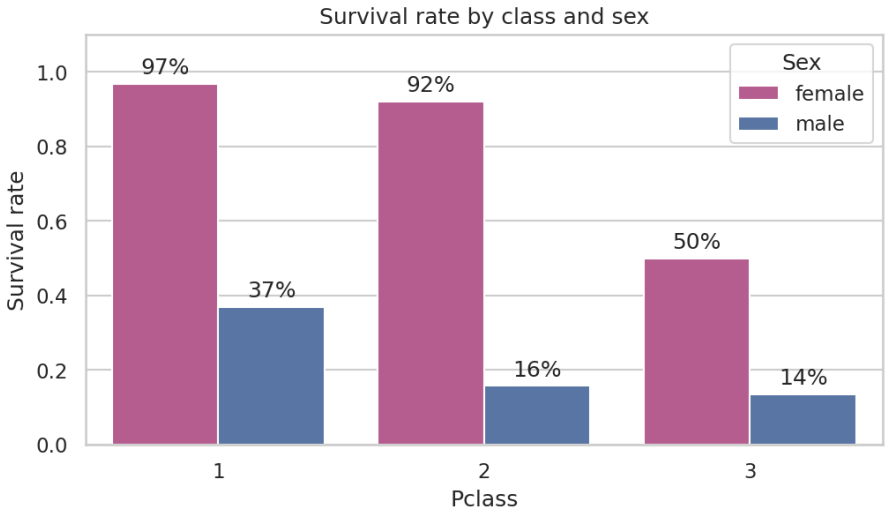
4.7 Scatterplot - Age vs Fare by outcome



Observations

- No linear relationship between Age and Fare (r = 0.10). The horizontal bands are fixed ticket prices paid by many passengers.
- Survivors (green) concentrate at higher fares: survival is 19.7% in the cheapest fare quartile and 58.1% in the most expensive one (69.5% for fares of 55 or more).
- The flat line of points at LogFare = 0 are the 15 zero-fare passengers (see anomalies).

4.8 Survival by Pclass and Sex together

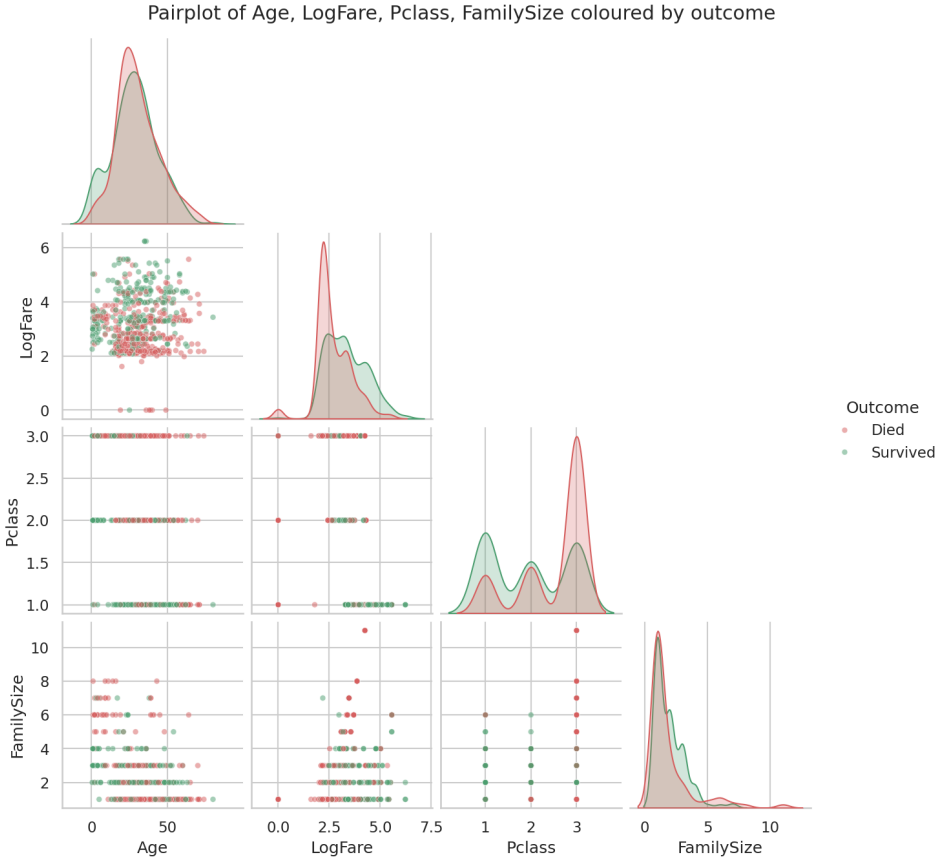


Observations

- **Within every class, women survived far more often than men**: 96.8% vs 36.9% (1st), 92.1% vs 15.7% (2nd), 50.0% vs 13.5% (3rd).
- Class also matters inside each sex: 3rd-class women (50.0%) did far worse than 1st/2nd-class women (over 92%).
- Sex outweighs class: 3rd-class women (50.0%) survived more often than 1st-class men (36.9%).

5. Multivariate analysis

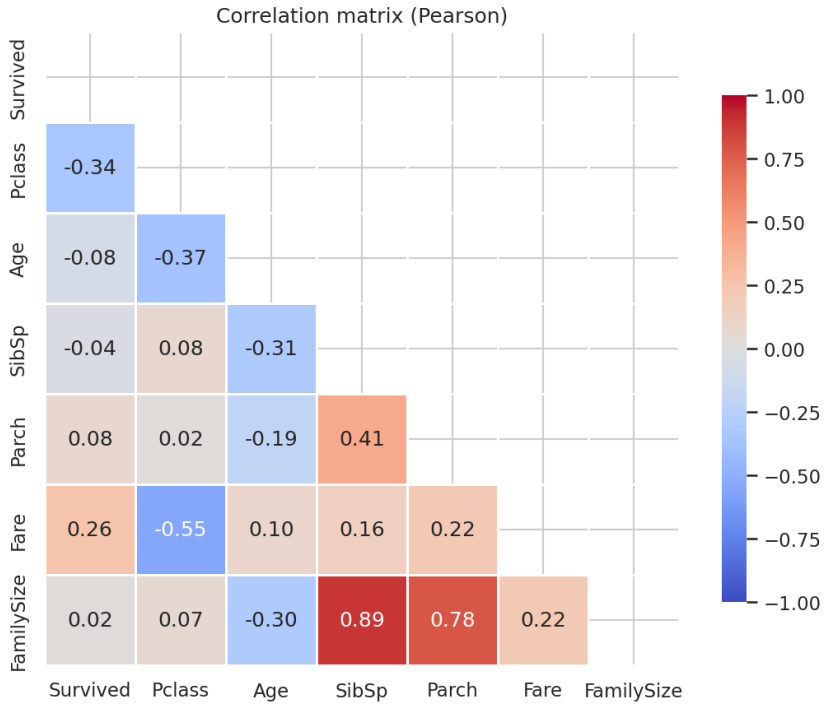
5.1 Pairplot (sns.pairplot)



Observations

- Built from Age, LogFare, Pclass and FamilySize for the 714 passengers with a known age. Survivors (green) sit at higher LogFare and better (lower) Pclass - that is where the two outcomes separate most clearly.
- Age vs FamilySize shows younger passengers travelling in larger families (correlation -0.30): children travelling with parents and siblings.
- Age vs LogFare shows no pattern beyond the class tiers, and the diagonal densities confirm the earlier findings (extra survivor bump for young children, survival odds differing by family size).

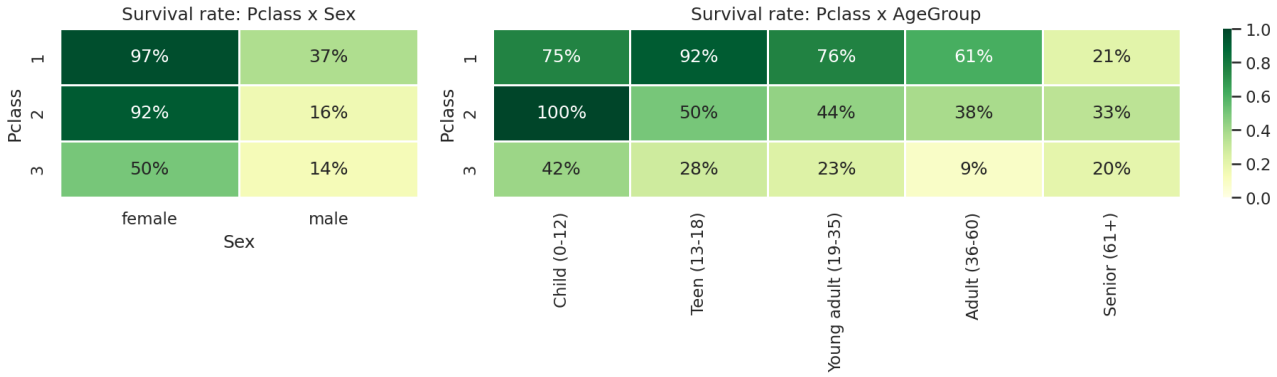
5.2 Correlation heatmap (sns.heatmap)



Observations

- **Links with Survived**: Pclass -0.34 and Fare +0.26 are the strongest; Age (-0.08), SibSp (-0.04) and Parch (+0.08) are weak linear signals. Sex is categorical so it is not in this matrix, even though it is the strongest factor.
- **Pclass-Fare -0.55 and Pclass-Age -0.37**: richer and older passengers sit in higher classes.
- **SibSp-FamilySize 0.89 and Parch-FamilySize 0.78**: FamilySize is built from SibSp + Parch, so these are redundant (multicollinearity) - use FamilySize or the two originals, not all three.

5.3 Survival-rate heatmaps



Observations

- **Left (Pclass x Sex)**: survival ranges from 97% for 1st-class women to 14% for 3rd-class men.
- **Right (Pclass x AgeGroup)**: every 2nd-class child survived (17 of 17), 3rd-class children 42%, 1st-class seniors only 21% (n=14) and 3rd-class adults (36-60) just 9% (n=58).
- Cells with fewer than about 20 passengers are noisy, so the extreme values should be read as hints, not facts.

6. Statistical checks

Test	Feature	Result
Chi-square vs Survived	Sex	chi2 = 260.7, p = 1.2e-58, Cramer's V = 0.54
Chi-square vs Survived	Pclass	chi2 = 102.9, p = 4.6e-23, Cramer's V = 0.34
Chi-square vs Survived	Embarked	chi2 = 26.0, p = 2.3e-06, Cramer's V = 0.17
Mann-Whitney U	Fare by Survived	p = 4.6e-22 (significant difference)
Mann-Whitney U	Age by Survived	p = 0.16 (no significant difference)
VIF (raw numeric)	Pclass / Fare / Age / SibSp / Parch	1.71 / 1.58 / 1.29 / 1.27 / 1.23

- **Chi-square tests**: Sex, Pclass and Embarked are all significantly associated with survival ($p < 0.001$). Effect size (Cramer's V) is largest for Sex (0.54), then Pclass (0.34), then Embarked (about 0.17).
- **Mann-Whitney U**: Fare differs significantly between survivors and non-survivors ($p < 0.001$) while Age does not ($p = 0.16$).
- **VIF** for the raw numeric features is below 2 (Pclass 1.71, Fare 1.58, Age 1.29, SibSp 1.27, Parch 1.23), so there is no serious multicollinearity. Adding FamilySize makes the correlation matrix numerically singular (condition number about 6e14) because it is an exact sum of SibSp and Parch.

7. Anomalies and data-quality flags

- **15 passengers paid a Fare of 0** - all men (5 in 1st, 6 in 2nd, 4 in 3rd class); only 1 survived. These may be complimentary tickets or missing values, so they are kept but flagged.
- **3 passengers paid 512.33**, the maximum, and all survived. These are extreme but real outliers.
- **7 infants under 1 year all survived**, while of 7 passengers aged 70 or older only 1 survived.
- **An 11-person family** (7 members in the data) - none survived.
- **Shared tickets**: 344 passengers share a ticket number with someone else (group travel), so the Fare column may reflect the whole ticket rather than a per-person price.
- **Missing Age is not random** (see data quality section): it is concentrated in 3rd class.

8. Summary of findings

1. **Overall**: 342 of 891 passengers survived (38.4%).
2. **Sex was the dominant factor**: 74.2% of women vs 18.9% of men survived; the pattern holds inside every class.
3. **Class mattered strongly**: 63.0% / 47.3% / 24.2% survival for 1st / 2nd / 3rd class. Fare, cabin availability and age all follow the same class ladder, so they overlap with Pclass.
4. **Women and children first is visible**: children aged 0-12 survived at 58.0%, and boys (Master) at 57.5% vs 15.7% for adult men.
5. **Family context**: passengers travelling alone (30.4%) and in large families of 5+ (16.1%) did worse than those in families of 2-4 (55-72%).
6. **Interactions are key**: 3rd-class women (50.0%) survived more often than 1st-class men (36.9%); 3rd-class adults had only 9% survival.
7. **Port of embarkation** shows differences (Cherbourg 55.4% vs Southampton 33.9%), but largely because of the class mix of each port.
8. **Data quality**: Cabin is 77% missing, Age 20% missing (not at random), Fare is heavily skewed (4.79 -> 0.39 after log transform) and there are 15 zero-fare passengers.
9. **Next steps for modelling**: use Sex, Pclass, Title, FamilySize, LogFare and HasCabin; impute Age by Pclass/Sex medians (or Title); keep only one of FamilySize or SibSp/Parch; try logistic regression and a decision tree as baselines.

9. Interview questions

1. What is EDA and why is it important?

Exploratory Data Analysis is the process of summarising and visualising a dataset to understand its structure, quality, patterns, relationships and anomalies before modelling. It catches missing values, outliers and wrong data types early, guides feature engineering and the choice of model, and prevents building on false assumptions. In this project EDA revealed that sex and class dominate survival, that Fare needs a log transform and that Cabin is too sparse to use directly.

2. Which plots do you use to check correlation?

A correlation heatmap for all numeric pairs at a glance; scatterplots (or regplot/jointplot) for a specific pair; pairplot for all pairs plus distributions. For numeric vs categorical use boxplots/violin plots; for two categorical variables use grouped bars, crosstabs and a chi-square test.

3. How do you handle skewed data?

First measure it (skewness, histogram). Then apply a transform such as log / log1p, square root, Box-Cox or Yeo-Johnson; cap extreme values (winsorising); use robust statistics like the median/IQR; or use tree-based models that do not care about skew. Example here: Fare skewness 4.79 became 0.39 after log1p.

4. How to detect multicollinearity?

Look at the correlation matrix/heatmap (|r| above about 0.8 is a warning), compute the Variance Inflation Factor (VIF above 5-10 is a concern) and check the condition number. Fix it by dropping or combining features, using PCA, or regularisation (Ridge/Lasso). Example here: FamilySize is an exact sum of SibSp and Parch (r = 0.89 and 0.78).

5. What are univariate, bivariate and multivariate analyses?

Univariate looks at one variable at a time (histogram of Age, countplot of Sex). Bivariate looks at the relationship between two variables (survival rate by Sex, Fare by Pclass). Multivariate looks at three or more together (pairplot, correlation heatmap, survival by Pclass x Sex x AgeGroup).

6. Difference between heatmap and pairplot?

A heatmap is a colour-coded matrix of values, usually the correlation coefficients (or a pivot table), so it is compact and quantifies only linear strength. A pairplot is a grid of scatterplots plus distributions for every pair of variables, so it shows shape, clusters, outliers and group separation, but gets crowded and slow with many variables. Use the heatmap to screen, the pairplot to inspect.

7. How do you summarise your insights?

Lead with the finding, back it with a number and a visual, explain why it matters, state caveats (sample size, confounding, missing data) and end with a recommendation. Example: 'Women survived at 74% vs 19% for men, and the gap holds in every class; sex and class should be the core model features.'